

## Homework 2: Relational Algebra

Name:

**Instructions:** Print this assignment single-sided. Each problem below is labeled with a “spiciness” rating, ranging from 🌶️ (mild), 🌶️ 🌶️ (medium), to 🌶️ 🌶️ 🌶️ (hard), according to how difficult I perceive the problem might be. You are welcome to use the back of each page to show your work. If you use any additional scratch paper, append it to the end of the assignment.

**Submission:** After you’ve filled in the answers, scan all pages into a PDF, and submit to canvas.

### Problems

1. Design a *schema diagram* for a company like Netflix that minimizes redundancy and NULL values. You should recall that an attribute cannot hold a list of values. Indicate all primary keys (underlined) and foreign keys (directed edges from the dependent relation to the source relation). (🌶️ 🌶️)

You need to store data on the following:

- A Movie has a title, description, country of origin, and a release-year. Assume that there may be two movies with the same name (e.g., Godzilla), but they are never released in the same year. You may assign each movie a unique identifier, but you shouldn’t need to.
- An Actor has a name and a biography. Assume that there may be two actors with the same name. You may assign each actor a unique identifier.
- A User has a userid and password. The company does not allow two different users to share the same userid.
- You need to keep track of what movies the users have seen.
- Users can rate movies from 1 to 5 stars. Note that they need not have seen the movie to rate it! A user can only rate a movie once (though that rating can change later).
- Users can also comment on each actor and/or movie. A comment has a date, time, and the comment itself. Unlike ratings, users can leave a multiple comments on the same movie or actor.
- Actors play certain roles in certain movies. Actors may also play multiple roles in the same movie (see: Eddie Murphy and Mike Myers).

Space for your schema diagram is given on the next page.



2. Let's define some new relational operations using a composition of various *primitive* relational and set operators! Because we already defined  $\bowtie$  operation using only primitive operators in class, you may safely use  $\bowtie$  in helping define the two operators below. For the problems below, assume we have relations  $R(A_1, \dots, A_n)$  and  $S(B_1, \dots, B_m)$ .

- (a) The Left Semi-Join  $R \ltimes S$  retains all tuples in  $R$  that participate in the natural join between  $R$  and  $S$ . For instance, Assume we have two relation schemas  $R(A, B)$  and  $S(B, C)$  and that that  $R = \{(1, 2), (2, 3), (3, 4)\}$ , and  $S = \{(2, 2), (2, 3), (5, 1)\}$ , then  $R \ltimes S$  produces  $\{(1, 2)\}$ . (👉)

I'll do this one:  $R \ltimes S = \Pi_{A_1, \dots, A_n}(R \bowtie S)$ . We start by producing the natural join, which by definition would return a table containing attributes  $\{A_1, \dots, A_n\} \cup \{B_1, \dots, B_m\}$  and the merging of tuples in  $R$  and  $S$  that agree on the common attributes. Next, we project only the attributes  $A_1, \dots, A_n$  to retain the tuples in  $R$  that participated in the join.

- (b) The Symmetric Difference  $R \triangle S$  returns the set of tuples found in either  $R$  or  $S$  but not both. For instance, the symmetric difference between  $R$  and  $S$  as defined in the previous problem gives us  $\{(1, 2), (3, 4), (2, 2), (5, 1)\}$ . (👉)

$R \triangle S = ?$

- (c) The Right Anti-Join  $R \Join S$  retains all tuples in  $S$  that do not participate in the natural join between  $R$  and  $S$ . For instance, the right-anti join between  $R$  and  $S$  as defined in the previous problem produces  $\{(5, 1)\}$ . (👉)

$R \Join S = ?$

For the remaining problems, consider the relation instances below that store data for college sports. Schools are split into two major conferences: A and B. Within each conference, there are four divisions: A-North, B-South, etc. In each game, there is a “home team” and an “away team”, identified by the schoolIDs. For instance, gameID 7 was between GaTech (away) and ASU (home) on September 8, 1980. In that game, GaTech won 24 – 6.

**School**

| schoolID | city          | school     | conference | division | spent |
|----------|---------------|------------|------------|----------|-------|
| 0        | Seattle       | Washington | B          | West     | 500M  |
| 1        | Cleveland     | Case       | A          | North    | 100M  |
| 2        | Pittsburgh    | CMU        | A          | North    | -50M  |
| 3        | San Francisco | Berkeley   | B          | West     | 800M  |
| 4        | Oakland       | Mills      | A          | West     | 350M  |
| 5        | Tempe         | ASU        | A          | West     | 400M  |
| 6        | Miami         | Miami      | A          | East     | 200M  |
| 7        | Houston       | Rice       | B          | South    | 250M  |
| 8        | New York      | Columbia   | A          | East     | 800M  |
| 9        | New York      | NYU        | A          | East     | 400M  |
| 10       | Buffalo       | SUNY       | B          | East     | 100M  |
| 11       | Atlanta       | GaTech     | A          | South    | 200M  |
| 12       | Baltimore     | JHU        | A          | North    | 0M    |
| 13       | Lafayette     | Purdue     | A          | South    | 90M   |

**Game**

| gameID | Away | Home | date  | year | awayScore | homeScore |
|--------|------|------|-------|------|-----------|-----------|
| 0      | 0    | 11   | 9/2   | 2012 | 0         | 3         |
| 1      | 3    | 13   | 10/17 | 2009 | 23        | 0         |
| 2      | 5    | 10   | 10/10 | 2012 | 10        | 10        |
| 3      | 4    | 9    | 11/20 | 2015 | 17        | 7         |
| 4      | 2    | 7    | 9/27  | 2014 | 7         | 14        |
| 5      | 9    | 8    | 10/30 | 1990 | 14        | 15        |
| 6      | 8    | 3    | 8/30  | 1980 | 21        | 9         |
| 7      | 11   | 5    | 9/8   | 1980 | 24        | 6         |
| 8      | 4    | 2    | 10/28 | 1981 | 35        | 15        |
| 9      | 12   | 10   | 11/27 | 2012 | 3         | 40        |
| ...    | ...  | ...  | ...   | ...  | ...       | ...       |

**Coach**

| name    | schoolID | title           |
|---------|----------|-----------------|
| Carroll | 0        | Head Coach      |
| Jackson | 1        | Head Coach      |
| Tomlin  | 2        | Head Coach      |
| Kelly   | 3        | Assistant Coach |
| Brown   | 1        | Head Coach      |
| Day     | 0        | Assistant Coach |
| ...     | ...      | ...             |

Answer each of the following queries using relational algebra. Below every prompt, I give you the results that your query should produce in blue. You are allowed “break up” a long-winded expression into smaller manageable for ease of writing. For example,

$$S \leftarrow \sigma_{city=Seattle}(School) \quad // \text{ } S \text{ stores all schools located in Seattle}$$

$$T \leftarrow \sigma_{(Away=S.SchoolID) \vee (Home=S.SchoolID)}(S \times Game) \quad // \text{ } T \text{ is all games in which the Seattle schools played in}$$

... //  $S$  and  $T$  can still be used later

3. Write a relational algebra expression that lists all schools (just the city and school name columns) in the Eastern and Northern divisions of Conference “A.” (🐼)

| city       | school |
|------------|--------|
| Cleveland  | Case   |
| Pittsburgh | CMU    |
| Oakland    | Mills  |
| Tempe      | ASU    |
| ...        | ...    |

4. List all 2015 games (gameID, awayScore, homeScore) in which the home team lost. (🐼)

| gameID | awayScore | homeScore |
|--------|-----------|-----------|
| 1      | 23        | 0         |
| 3      | 17        | 7         |
| 6      | 21        | 9         |
| 7      | 24        | 6         |
| 8      | 35        | 15        |
| ...    | ...       | ...       |

5. List all the schools for which ‘Kelly’ has worked as an Assistant Coach. Assume you do not know the rest of the data, so you can’t simply select schoolID 3 out of the School table. You need to join Coach with School... (🐼)

| school   |
|----------|
| Berkeley |
| ...      |

6. List all games (gameID, date, year) in which Columbia University participated. Assume you do not know Columbia's schoolID off-hand, so you must first find it in the School table. ( 🐼 )

| gameID | date  | year |
|--------|-------|------|
| 5      | 10/30 | 1990 |
| 6      | 8/30  | 1980 |
| ...    | ...   | ...  |

7. How many of each of type of coach does every school employ? In the data that is listed, University of Washington has 1 Head Coach (Caroll) and 1 Assistant Coach (Day). Your result should contain three columns: the school, the coach title, and the number of coaches having that title. ( 🐼 🐼 )

| school     | title           |     |
|------------|-----------------|-----|
| Washington | Head Coach      | 1   |
| Washington | Assistant Coach | 1   |
| Case       | Head Coach      | 2   |
| CMU        | Head Coach      | 1   |
| Berkeley   | Assistant Coach | 1   |
| ...        | ...             | ... |

8. The league commissioner suspects that conference B may be taking in improper funds. Return a list of schools (city, school name, expenditure, avgOfConfB) *outside* of conference B that spent less than the average spent by all schools in conference B. A couple of pieces make this one slightly trickier: (1) You need to use aggregation to get the conference B expenditure, and (2) How can you use < to compare with the average you extracted, when < is not a relational operator? ( 🐼 🐼 )

| city       | school | spent | avgOfConfB |
|------------|--------|-------|------------|
| Cleveland  | Case   | 100M  | 412.5M     |
| Pittsburgh | CMU    | -50M  | 412.5M     |
| Oakland    | Mills  | 350M  | 412.5M     |
| Tempe      | ASU    | 400M  | 412.5M     |
| ...        | ...    | ...   | ...        |

9. How many total points (away score and home score combined) were scored in the 2023 season? There are a few caveats: (a) There is no  $+$  operator in relational algebra, but there is a `sum()` aggregation function. (b) Duplicate tuples are removed, which can screw up your sum! (3) When applying a set operator (e.g., a union), you must assume that the result of the union is unnamed. You should get a relation storing a single total, as shown below (the value 728 is made up since we don't know the rest of the data.) ( 🐾 🐾 🐾 )

| total |
|-------|
| 728   |

10. Which year(s) did the school 'Case' record its fewest wins in its history? Output the year and the win count. Note that there may be multiple years during which 'Case' recorded their lowest win total. Do not assume you know what Case's schoolID is off-hand. The table below is just an example of what you should see. (*This one may be a bit long – I am so sorry LOL* 🐾 🐾 🐾 )

| year | numWins |
|------|---------|
| 2001 | 2       |
| 1997 | 2       |
| ...  | ...     |